

Design Studies
A Reader
Edited by
Hazel Clark
David Brody

DESIGNING ENGINEERS

Louis Bucciarelli

DO YOU KNOW HOW YOUR TELEPHONE WORKS?

A few years ago, I attended a national conference on technological literacy, a topic that had recently attracted attention in the media as well as among scholars concerned with education at both secondary and postsecondary levels in the United States. One of the main speakers, a sociologist, presented data he had gathered in the form of responses to a questionnaire. After detailed statistical analysis, he had concluded that we are a nation of technological illiterates. As an example, he noted how few of us (less than 20 percent) know how our telephone works.

This statement brought me up short. I found my mind drifting and filling with anxiety. Did I know how my telephone works?

I squirmed in my seat, doodled some, then asked myself, What does it mean to know how a telephone works? Does it mean knowing how to dial a local or long-distance number? Certainly I knew that much, but this did not seem to be the issue here. Might it mean knowing how to install a phone, or perhaps knowing what to do when something goes wrong—no dial tone, noise on the line? But most of us do know that in such cases we need only call a service department. No, I suspected the question was to be understood at another level, as probing the respondent's knowledge of what we might call the "physics of the device."

I called to mind an image of a diaphragm, excited by the pressure variations of speaking,

vibrating and driving a coil back and forth within a magnetic field, generating a variation in voltage across the terminals of the coil. I could even picture this signal traveling along the wire coming out of the phone into the wall and out onto the pole outside my house. But what then? How do all those different signals from different phones get sorted and routed to their proper destinations? I had no explanation for this.

If this was what the speaker meant, then, he was right: Most of us don't know how our telephone works, and I bet if he asked, most of us don't give a damn about knowing how it works as long as it does. Despair and guilt had yielded to cynicism.

Indeed, I wondered, does he know how his telephone works? Does he know about the heuristics used to achieve optimum routing for long-distance calls? Does he know about the intricacies of the algorithms used for echo and noise suppression? Does he know how a signal is transmitted to and retrieved from a satellite in orbit? Does he know how AT&T, MCI, and the local phone companies are able to use the same network simultaneously? Does he know how many operators are needed to keep the system working, or what those repair people actually do when they climb a telephone pole? Does he know about corporate financing, capital investment strategies, or the role of regulation in the functioning of this expansive and sophisticated communication system?¹

Does anyone know how their telephone works?

How about the designer of the device? Surely, he or she must know. I followed this line of thought

for a while, but again, I found my footing less sure than I had anticipated. Indeed, from my own observations, I can claim fairly confidently that there is no single individual alone who knows how all the ingredients that constitute a telephone system work together to keep each of our phones functioning. There is no one “maker.” Instead, inside each firm, there are different interests, perspectives, and responsibilities—corporate planning, engineering, research, production, marketing, servicing, managing—and consequently different ways in which the telephone “works.”

At this point I retreated from cynicism. The question now struck me as interesting, not as an instrument for testing technological literacy, but in its own right: What does it mean when someone claims to know how their telephone works? I conjectured that there could well be no unique criterion for judging responses; there could be as many legitimate, that is to say accurate, ways to describe how the telephone works as there are respondents.²

The narrow view of the workings of the telephone has the quality of a myth. But while the story about vibrating diaphragms and coils moving in a magnetic field may provide coherence amid complexity, may give us the confidence to respond in the affirmative to a sociologist on the track of technological literacy, and may encapsulate some particular facet of technical truth, taking it as the measure of a person’s understanding of how today’s sophisticated and dynamic system of communication works is naive.

This naivete is just one failing of the sociologist’s research program; another is the presumption that you can test a person’s “technological literacy” by literary means alone. Thus, if a respondent has just finished reading about the telephone in the *Encyclopedia of Science*, the sociologist would no doubt find him or her technologically literate. But this is neither sufficient nor necessary as a display of knowing how the telephone works. Indeed, it is little more than a test of reading retention.

No, the “knowing how it works” that has meaning and significance is knowing how to do

something with the telephone—how to act on it and react to it, how to engage and appropriate the technology according to one’s needs and responsibilities. Thus the everyday user around the house from the age of seven to seventy, the person who repairs the lines, the new product engineer, and the corporate executive each appropriates the telephone to his or her own interests, and each knows how it works.

Refocusing in this way, dragging myself away from an instrumental story derived from a vision of the scientific origins of the device, led me to reformulate the question. Rather than “Do you know how your telephone works?” I would ask, “How does the telephone symbolize, alienate, serve, or have meaning for you?” With this question we might be able to test the breadth and depth of a person’s technological understanding and competence. Note, too, how, with this reorientation, the object loses its grip and the social context of its workings assumes primacy.

It is the fixation on the physics of a device that promotes the object as an icon in the design process. For while different participants in design have different interests, different responsibilities, and different technical specialties, it is the object as they see and work with it that patterns their thought and practice, not just when they must engage the physics of the device but throughout the entire design process, permeating all exchange and discourse within the subculture of the firm. This way of thinking is so prevalent within contemporary design that I have given it a label—“object-world” thinking—and will now detour a bit to give a preview of what I mean by the term. I will start with a less complex and more ancient technology and explore what it might mean to know how it works from the perspective of an object world.

HOW A CHAIR WORKS—ONE PERSPECTIVE

Clearly we all know how to sit in a chair, so in this respect we know how it works: A chair is for sitting. That is how it functions. Unfortunately,

this description is akin to saying that a telephone is for making telephone calls. Surely we can do better. (You never know when a sociologist is going to telephone to ask if you know how your chair works.) How about this?

A chair is a supported seat. A seat is a two-dimensional, finite, contoured surface, made of a rigid or flexible material, which provides immediate support to at least the buttocks of the human body. This support is sufficient to enable a person, when seated, to lift both feet off the ground without falling over. The surface is supported by a structure intervening with the ground.

A skeptic might interject here, “Does this mean that a hammock is a chair? Or a bed? Or a horse?” In response, we can start adding details. We might, for example, say that a chair is made of inanimate material. There goes the horse.³ Or we can pass the buck back to a simpler precedent and build upon that:

A chair is a stool with a backrest, and a stool is a board elevated from the ground by supports.⁴

But this is still not good enough, for we still have to explain how a stool works. Why do most stools have three legs while a chair has four? We must strive to be even more scientific:

A chair is a four-legged stool with a back. Although three legs, or points of support, are sufficient to support the seat, the stability of seating is enhanced if we add a fourth point of support since, for said stability, a line connecting the center of gravity of an extended body (yours or mine) to the center of gravity of the earth (or whatever planetary body one happens to be seated upon) must intersect the plane through the feet of the legs of the stool or chair, at a point interior to the three- or four-sided plane figure having the feet as vertices. With four legs, one can assure stability with a square seat.⁵

This, then, is a chair and how it works. We can call this the principle of operation of the chair—what Robert Pirsig would call its underlying form.⁶ This is the “physics of the device” knowledge that is often taken as the hallmark of technological literacy. It constitutes a sparse, efficient, generic, abstract identity of a chair, just as the identity of a circle is defined by the Platonic ideal of “all points equidistant from a fixed point.”⁷

Given this principle, we can imagine an infinite variety of particular embodiments that would function as a chair. A chair can be finely carved or sparse yet stately. We can add arms and cushions to create a throne—or set the thing up on curved rails to make a rocker—a matter of controlled stability—if we want to get frivolous.

I don’t claim that this generic description is the best possible in any sense. Nor is it unique. Indeed, there are other object worlds we should enter to define more fully how a chair works. We can talk about the forces borne by the legs, which distort and deform relative to the seat when a person sits down. We can even do a finite-element analysis on the computer and estimate the internal stresses in the legs, the seat, and the back. And we certainly ought to describe the craft knowledge of chair object worlds: how to join the legs and back to the seat, whether or not it is preferable to make the back an extension of the rear two legs. All of this is part of an object world perspective on how a chair works.

HISTORICAL ROOTS—OBJECT WORLDS

But this is neither a physics text nor a chair design manual. My intent instead is to illustrate a particular way of thinking about and framing an answer to the question “How does it work?” This way of thinking will prove essential to understanding the thought and practice of participants in contemporary engineering design.

This has not always been the case. The first chair makers did not need to enter the world of geometry, consider the conditions that ensure

static equilibrium, or call upon mathematical theories describing stress and strain within beams, legs, and arms in order to do their work. Vernacular technique was all they needed. One might claim that the underlying form of the chair was embedded in the crafter's rules of thumb, sense of symmetry, feel for an awl in making a cut, or know-how about joining wood to wood, but that knowledge remained unarticulated, tacit, and sensual (as far as we know). It was not of a kind with contemporary scientific understanding—for example, a Ptolemaic astronomer's explanation of planetary orbits, which relied upon the abstract notion of uniform circular motion and the machinery of epicycles and deferents (the elements of underlying form of the motion of the planets) and was drawn out in texts for all to read who could. But then, until relatively recently, the heavens were of a different nature than the stuff of the sublunary region. Indeed this “underlying form” way of thinking about earthly, mundane things like chairs and telephones was very much a child of the Renaissance. With the shattering of the perfection of the heavens, with the connection of the terrestrial to the celestial (the earth is not below, on center; the sun has spots; the planets are not wandering stars and themselves have moons; our moon is mountainous and, like Newton's apple, is falling toward the earth), the decaying stuff of this world became a candidate for abstract philosophical thought and discourse. Some attribute to Galileo this new way of framing and responding to the question.

Image 6 shows Galileo's cantilever beam.⁸ He wanted to know how it works in object-world terms. That is, he was seeking, and felt he had found, its underlying form. He made the following claim:

A solid prism or cylinder of glass, steel, wood, or other material capable of fracture, which suspended lengthwise will sustain a very heavy weight attached to it, will sometimes be broken across ... by a very much smaller weight, according as its length exceeds its thickness.⁹

Already we note the generic nature of his vision of “glass, steel, wood, or other material capable of fracture” and the instrumental relationship, somewhat hidden to the modern eye, in the proposition “be broken across ... by a very much smaller weight according as its length exceeds its thickness.” His demonstration immediately follows:

Let us imagine the solid prism ABCD fixed into a wall at the part AB; and at the other end is understood to be the force of the weight E. (Assuming always that the wall is vertical and the prism or cylinder is fixed into the wall at right angles.) It is evident that if it must break, it will break at the place B, where the niche in the wall serves as support, BC being the arm of the lever on which the force is applied. The thickness BA of the solid is the other arm of this lever, wherein resides the resistance, ... the moment of the force applied at C has, to the moment of the resistance which exists in the thickness of the prism (that is, in the attachment of the base BA with its contiguous part), the same ratio that the length CB has to one-half of BA. Hence the absolute resistance to fracture in the prism BD (being that which it makes against being pulled apart lengthwise) ... has, to the resistance against breakage by means of the lever BC, the same ratio as that of the length BC to one-half of AB And let this be our first proposition.

The figure is striking—a fine etching in the Renaissance genre of nature observed. But it is also disconcerting. There is a lack of proportion in the weight hooked to the end of the beam relative to the beam itself. And the wall looks like the decaying remnant of some edifice that was perhaps once grand. How is it that the structure remains intact, supporting this massive weight, while the wall at the root of the beam seems fallen to such a sorry state? And the letters of the alphabet—what are they doing in this landscape?

The image is not to be read in this naturalistic way. No, it is a new kind of figure, a new image



Image 6 Telephone: *Model 302*. Designed by Henry Dreyfuss (American, 1904–1972). Manufactured by Western Electric Manufacturing Company for the Bell Telephone Company, ca. 1937. Metal, paper, rubber-sheathed cord. Cooper-Hewitt, National Design Museum, Smithsonian Institution; museum purchase with the Decorative Arts Association Acquisition Fund, 1994-73-2. Photo: Hiro Ihara.

to go with a new science. Galileo means to help the reader follow his argument—that the beam will fail at the end B if the weight at the end C becomes too large. But there is more to it than that. The reader must see in the figure an angular lever with its fulcrum at B, one arm extending out along the beam to the weight at the end C, the other, shorter arm, extending from B up to A. Then the reader must imagine an internal force, a “resistance” acting over the arm AB, directed to the left, and visualize how this resistance would balance the weight suspended at E through the action of the lever. From the principle of equilibrium of the lever one obtains that the ratio of the force that will break the beam when applied transversely at E to the force that will break the beam when applied longitudinally at AB—a

force Galileo calls “the prism’s absolute resistance to fracture”—is the same as the ratio of one-half the distance AB to the distance BC.

This is how we are meant to read Galileo’s proposition. He is explaining how a cantilever beam works by revealing its underlying form. His explanation is not a proof, though. The law of the lever was already well known. What is significant and innovative is the association of the workings of a load-bearing beam with the geometry and principle of the lever. The new science comes in positing a relation between the beam as physical artifact and vernacular technique and the beam as abstract lever and mathematical artifice.

As a *physical artifact*, the beam has a “thickness,” a “length,” a “mortise,” and a “weight” at the end. These are significant, but the embellishments of

these characteristics—the crumbling nature of the wall, the ivy, the grain in the wood, even the material from which the beam is made—are not.

As an *abstract lever*, there is a fulcrum at B, a force applied at C, and an (internal) resistance that “opposes the separation of the part BD lying outside the wall from that portion lying inside,” and they bear the same ratio as half the length (lever arm) BA to the length (lever arm) BC. That is how the beam works.

Indeed, that is a cantilever beam. With Galileo, the questions “What is it?” and “How does it work?” collapse into one. When put, they solicit the same response. The attempt to answer either brings out the fact that a cantilever beam is a lever, just as a chair can be seen as a stable platform. In this, “we take for true being that which is only a method.”¹⁰

WHO DEFINES THE WORKINGS?

Seeking underlying form is but one approach to defining how a telephone, a chair, or a cantilever beam works. It guides participants designing within object worlds but doesn’t always make sense for those of us who don’t engage technology in this intimate way. Indeed, it is not always relevant, or all there is to the workings of technique; and sometimes it sounds downright silly to speak of a chair in this way.

For example, as you stand before Napoleon’s throne in the Louvre, you are probably not thinking about underlying form. Or if you encounter a Rietveld chair in the Centre Pompidou, with its bright red back, glossy black seat, and its spindly arms, legs, and interlacing struts all tipped with yellow, you might think about underlying form only if it crosses your mind to wonder whether or not it would collapse if you sat on it. Or if you spot a Le Corbusier lounge chair in the window of a fashionable shop along the Rue Rivoli, priced at 12,380 francs, its inner constitution is not likely to be the first thing that passes through your mind. You don’t spend \$2,000 on a four-legged stool with a back.

A Rietveld or Corbusier chair or Napoleon’s throne “works” as something other than what you or I am seated on at the moment. They work as art, as fashion, as symbol of power and prestige. Their primary function is not to support bodies. Their one common characteristic is that they are uncommon. Each has its own special identity that renders irrelevant our generic and abstract description. Our object-world definition of chair is meaningful only if we are talking about an anonymous chair. The anonymity of this chair is of a piece with the generic and abstract response we give to the question of how it works. It is no coincidence that Galileo’s vision takes root at the first glimmerings of an industrial revolution.¹¹

Far removed from the Renaissance, most of us don’t see technology in terms of underlying form. We live in a world where each object appears in its particularity, as an artifact made of hardware or as a system of rules and constraints. The underlying form remains hidden, sometimes intentionally; our perceptions of how a telephone, a chair, a photovoltaic module, or an x-ray inspection machine function can be correspondingly “superficial.” Yet we do know how they work, as shown by our knowing how to pass them by, work through them, remake them, or use them, even if but briefly.

Not all relations are superficial. Often the connection is dictated. A worker in a machine shop who tends a semiautomatic tool, which requires the insertion, then removal of the workpiece in but a second or two, has little choice in how to relate to technique. Other relationships are direct and intense by choice. Bricolage and do-it-yourself are possible, as Pirsig has so beautifully demonstrated in *Zen and the Art of Motorcycle Maintenance*. For example, an owner’s careful and constant ministrations might cause a motorcycle or automobile to reveal a personality; or an office worker might become enthused about “hacking,” learning to navigate the computer system in a personal and deliberate way. Moreover, even an ephemeral connection can be quite intense, as when a family reacts to the siting of a recycling

facility across the town field; they may never have seen one, but they still have enough of a sense of how one might work to want to know more before they go along with the idea.

The way in which one sees how technology works is very much a matter of the nature of the encounter—whether it is in passing, intense in bricolage or dictation, or lay-political. Our relations to and hence our perspectives on technology may vary, but in general, as user, traveler, player, viewer, or tender, we do not have the same connection to technology that its makers have. However we appropriate technique, most of us do not see technology in terms of its formal structure, underlying form, or *inner constitution*.¹²

Those who are concerned with technological literacy often seem to be claiming that this is a sad state of affairs. Others see a deeper problem in our relations with technology, one that won't necessarily be solved by a universal program of education. They claim that we have embraced a new brand of slavery as we accept subservience to technique over which we have little control, much less understanding. Or, to put this idea in terms of a less odious metaphor, technology now legislates and structures our lives, yet we seem not to have been given the opportunity to vote.¹³

How does your telephone work? How does your chair really function? These questions fade in significance next to those that ask about legislating and ruling lives. Even if we can answer the sociologist "correctly" and thus be counted among the technologically literate, we remain uncomfortable: What disturbs us is the thought that, whatever our level of technological literacy, we are powerless to affect the form and function of the machinery that surrounds us, aids and comforts us, alienates and confuses us, improves our productivity, makes too much noise, sometimes leaks, can be beautiful, can be ugly, is often indispensable There it is: the telephone rings, and we answer.

It is a small step from these thoughts to a grand vision of technology that is autonomous in its workings, running our lives and out of the control of not just us ordinary citizens but engineers,

managers, and corporate strategists who themselves don't fully understand what they are doing.

NOTES

Extracted from L. Bucciarelli, *Designing Engineers* (Cambridge, Mass.: MIT Press, 1994), pp. 2–12. © 1994 Massachusetts Institute of Technology, by permission of the MIT Press.

1. "Only 19 percent believe that they have a clear understanding of how their telephone works." From Jon D. Miller, *Technological Literacy: Some Concepts and Measures*, Public Opinion Laboratory, Northern Illinois University, 19 February 1986. Note the inferred wording of the question. In this form, it is as much a test of a consumer's confidence in his or her technical understanding as it is a test of the latter. The qualifier "clear" no doubt assures a low positive response. Evidently, we will never know what respondents do know about how their telephone works. But note that, with the question phrased in this way, it is not necessary for the sociologist to know the slightest thing about how a telephone works in order to complete his research. I assume that a graduate student did the dialing.
2. This is not to claim that "anything goes"—that is, that some respondents will not offer erroneous accounts in their attempts to please a sociologist. Nor should one conclude that the telephone can be put to use for any arbitrary purpose or that the telephone is value free or neutral. It is a long way from the claim that different persons have different relationships with one and the same technology, the claim made here, to the fantasy that depicts technique as the sterile embodiment of nature's laws, devoid of human intent and interests.
3. The exclusion of ludicrous but rational and logically possible readings of statements and claims from the serious business of design is a matter of context. The neophyte engineer's

- suggestion will often be greeted with a laugh, and he or she learns, in this way, that "a horse is not a chair around here."
4. Christopher Dresser, *Principles of Decorative Design* (London; New York: Cassell, Peter & Galpin, 1873), quoted in E. Russell, P. Garner, and J. Read, *A Century of Chair Design* (Chicago: Academy Editions, 1980).
 5. The knowledge that three points of support are all that are required to support a seat (and to define a plane) is reflected in ancient pottery vessels. Those pots and tankards showing four legs generally mimic some form of animal life. The problem with four legs is that you had better have a level surface to set your vessel down upon or to place your chair on if you don't want the contents sloshing about or yourself rocking in an uncontrolled way.
 6. Robert Pirsig, *Zen and the Art of Motorcycle Maintenance* (New York: William Morrow, 1974).
 7. Note how some sort of abstract sketch or figure would be extremely useful for describing the chair's principle of operation. Not a glossy picture of a Le Corbusier chaise lounge, but a simple, geometrical figure with a few lines for legs, the outline of the plane figure whose vertices are located at the four points where the feet hit level ground, a special symbol for center of gravity, and finally a line drawn from the center of gravity, through the plane at the base of the feet, directed toward the earth's center. For stability, the intersection of this line with the plane at the feet must lie within the interior of the plane figure.
 8. For this image, see http://www.lindahall.org/events_exhib/exhibit/exhibits/civil/strength.shtml
 9. Galilei Galileo, *Discorsi e dimastrazioni matematiche, intorno a due nuoue scienze* (Leida: Appresso gli Elsevivii, 1638). For a standard translation, see *Dialogues Concerning Two New Sciences*, trans. Henry Crew. Alfonso de Salvio (New York: Macmillan, 1914), p. 116 (fig. 17).
 10. Husserl attributes to Galileo the conception of nature as, in reality, possessing a mathematical structure behind the appearances. The full Husserl quote is "The *Ideenkleid* (the ideational veil) of mathematics and mathematical physics represents and (at the same time) disguises the empirical reality and leads us to take for True Being that which is only a method." Quoted in Robert Cohen and Marx Wartofsky, eds., *Boston Studies in the Philosophy of Science*, vol. 2 (New York: Humanities Press, 1965), p. 286. Or, as put by Aron Gurwitsch in his "Comments on Marcuse" in the same volume (p. 293), "The world is not in reality as it looks."

See also Michael Lynch, "The Externalized Retina: Selection and Mathematization in the Visual Documentation of Objects in the Life Sciences," in Michael Lynch and Steve Woolgar, eds., *Representation in Scientific Practice* (Cambridge, MA: MIT Press, 1990), pp. 153–186, especially pp. 169ff.
 11. Even anonymity is not enough to ensure that we have an appropriate response to the question. Take an electric chair. Clearly, this is not particular in the "one-of-a-kind," museum-item sense (although no doubt some wax museum has one on display). How does an electric chair work? What is an electric chair? Our abstract way of answering would have to move beyond Galilean mechanics and refer to voltages and currents, resistive paths through the human body as well as the location of its center of gravity. But, again, we don't quite have it right.
 12. Ian Hacking, *Representing and Intervening* (Cambridge: Cambridge University Press, 1983).
 13. This line of criticism is forcefully presented in Langdon Winner, *The Whale and the Reactor* (Chicago: University of Chicago Press, 1986).